Sound speed and resonance

antocciphysics | TEACHER KEY | CPA + MATCHING SEI

Learning target: Use travel time and resonant conditions to interpret sound.

Accept equivalent correct methods and clearly labeled models. Language supports do not change the physics demand. Supplied data are examples; actual lab measurements and teacher checkpoints must be recorded from the lesson.

01. Warm-up: bring back an earlier idea *Review: Unit 6, Day 6.*
The 5 cm wave generally spreads more because wavelength is larger relative to the same opening.

02. Worked example: explain the model
Half a wavelength fits; displacement node near the middle and antinodes at open ends.

03. Read and interpret the model
Sound travels out and back; total 2×range.

04. Apply the physics
40 m total; 20 m range.

05. Explain your reasoning
Natural modes set preferred frequencies; response also depends coupling and damping.

06. Review: connect an earlier idea *Review: Unit 5, Day 9.*
30% efficiency; 420 J rejected, assuming a complete cycle.

07. Exit check
30 m.

08. STRETCH: test the model
340 and 170 Hz; doubling length halves fundamental frequency.

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01. Warm-up: bring back an earlier idea *Review: Unit 6, Day 6.*
Wavelength becomes 1/3; typically less spreading when the opening is large relative to the new wavelength.

02. Worked example: explain the model
Half a wavelength fits; displacement node near the middle and antinodes at open ends.

03. Read and interpret the model
Sound travels out and back; total 2×range.

04. Apply the physics
51 m;λ1 m; f 340 Hz.

05. Explain your reasoning
Natural modes set preferred frequencies; response also depends coupling and damping.

06. Review: connect an earlier idea *Review: Unit 5, Day 9.*
Work 1050 J; efficiency 30%.

07. Exit check
340 Hz.

08. STRETCH: test the model
Fundamental 100 Hz; odd harmonics 100,300,500 Hz.